# Venkata Lakshmi Parimala Pasupuleti

AI Developer · Custom MCP Tooling · Multi-Agent Orchestration · LLM Integration · Agentic Applications · ReAct · Responsible AI · Python · JavaScript · REST APIs · Generative AI

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## PROFESSIONAL SUMMARY

AI developer specializing in **custom MCP tooling, multi-agent orchestration, and LLM integration** - with 8+ years of enterprise software engineering depth. Built **AgentProbe**: a custom MCP-architecture generative AI evaluation platform with multi-provider LLM routing (OpenAI, Anthropic, Groq, Ollama), 5-metric eval engine, and CI-integrated regression gate - any agent connects via REST API without code changes. Built **Leasel**: 8-agent LangGraph agentic application applying **ReAct-style reasoning, Chain-of-Thought evidence synthesis, HITL escalation**, and responsible AI enforced in code - LLM reasons, deterministic logic makes every regulated call. At Entrata: **enterprise-scale Python/JavaScript/REST/JSON integrations** serving 100K+ MAU, RabbitMQ async workflows, AWS S3 pipelines, across 20+ production modules (120+ Agile sprints). M.S. Data Analytics Engineering, George Mason University (GPA 3.87).

## WORK EXPERIENCE

### Magizhs Tech Solution
May 2025 – Aug 2025

**AI/ML Engineer Intern — Agentic AI · LangGraph · Responsible AI · RAG**

- Built an **agentic AI application on LangGraph** (parse → plan → retrieve → validate → conditional routing) integrating LLM reasoning with deterministic guardrails - applied **Chain-of-Thought planning** to translate natural-language requests into structured workflows with **RAG-based grounding** over domain regulations; implemented intelligent automation from prompt to output with zero hallucinated compliance decisions.
- Enforced **responsible AI** in code: every compliance decision made by deterministic validation logic, not the LLM - blocking non-compliant actions before export regardless of LLM confidence; validated end-to-end **workflow automation** with human-in-the-loop review before final output delivery.

### George Mason University
Aug 2024 – May 2026

**Research Associate / Graduate Teaching Assistant — AI Curriculum · Statistical Modeling · Analytics**

- Built multi-source ETL pipelines and statistical models (Python, R, 190+ countries) - **$R^2 = 0.977$**; applied Isolation Forest anomaly detection; produced Tableau and ggplot2 dashboards driving 40% improvement in research comprehension for faculty leadership. Delivered technical workshops to **600+ students** covering Python, data structures, OOP, and AI system design.

### Entrata — Real Estate PropTech SaaS
Sep 2018 – Aug 2024 · Remote

**Senior Software Engineer — Enterprise Integrations · Python · JavaScript · REST/JSON · AWS · Workflow Automation**

- Designed and owned **enterprise-scale REST API + JSON integrations, RabbitMQ async messaging workflows, and AWS S3 document pipelines** across 20+ production modules serving **100K+ monthly active users**; all integrations owned through design, build, deployment, and production incident response across 120+ Agile sprints.
- Shipped **Python/JavaScript/PHP/TypeScript production systems**; led microservices redesign with Docker and CI/CD pipelines (50% throughput improvement, 30% deployment reduction); built **intelligent automation pipelines** transforming federal HUD source data into ~180K domain records powering eligibility and compliance decisions - retry logic and idempotency on async processing eliminating double-processing failures.
- **Technical documentation author** across 120+ sprints; **300+ peer code reviews; mentored 5+ engineers**; built internal debugging tools reducing debugging time 40% and keeping open bugs below 10 for 6+ months; led PHI-compliant AWS S3 migration (Libsodium encryption, zero security incidents).

### Entrata India (Db Xento Systems)
Jan 2017 – Sep 2018 · Pune

**Associate / Trainee Software Engineer — Full-Stack · REST APIs · Data Pipelines**

- Built REST API integrations and data workflows (PHP, Python, PostgreSQL, JavaScript) for property management and compliance; led AWS S3 migration; designed SQL import pipelines for 1.6M+ federal housing records.

## PROJECTS

### AgentProbe: Custom MCP Tooling & Generative AI Evaluation Framework
MCP · LLM EVAL

Python · MCP (Model Context Protocol) · LangGraph · FastAPI · ChromaDB · OpenAI / Anthropic / Groq / Ollama · React · CI/CD (GitHub Actions) — github.com/venkatalakshmiparimala/AgentProbe

- **Custom MCP-architecture AI evaluation platform**: 15-endpoint FastAPI backend, **multi-provider LLM integration** (OpenAI, Anthropic, Groq, Ollama) via unified routing layer, 5-metric eval engine (correctness, groundedness, latency, token efficiency, hallucination rate), CI-integrated regression gate blocking regressions pre-merge; any agent integrates via REST/JSON without code changes - plug-and-play enterprise connectivity. React observability dashboard tracks per-run metrics, retrieval drift, and latency. Validated across 1,240+ test cases: pass@1 81.7%, pass@3 88.3%.

### Leasel: 8-Agent Agentic Application — ReAct Reasoning · HITL · Responsible AI
INTELLIGENT AUTOMATION

Python · LangGraph (8-agent StateGraph) · FastAPI · Streamlit · RAG · ChromaDB · HUD API · OCR · SQLite — github.com/venkatalakshmiparimala/Lease_complaince_rag

- **8-node LangGraph orchestration** (Triage, Retrieval, Configuration, Historical Incident, Structured Data, Reasoning, Validation, Human Escalation) applying **ReAct-style perception-reasoning-action loops and Chain-of-Thought evidence synthesis** to automate complex compliance workflows; **HITL escalation** when confidence is insufficient; live HUD regulatory API + RAG over compliance documents; **responsible AI enforced in code**: LLM reasons, deterministic logic makes every regulated call. Results: 93.6% precision, 100% citation correctness, 0% unsupported-claim rate on 1,240 documents.

### AI on FHIR: NLP-Powered Healthcare Enterprise Integration
ENTERPRISE API

Flask · React · BERT · LLMs · FHIR REST API · Docker · spaCy · JavaScript · Embeddings — github.com/venkatalakshmiparimala/AI_on_FHIR

- Flask + React AI platform integrating **LLM-powered NLP with third-party FHIR REST/JSON API** - translating natural-language queries (EN/ES) into structured API calls using BERT and embeddings; ~60% reduction in manual clinical retrieval time; HIPAA-compliant RBAC; containerized with Docker. Demonstrates **intelligent automation applied to enterprise API integration**.

## EDUCATION

**M.S. Data Analytics Engineering** — George Mason University
May 2026 · GPA 3.87

AI systems, ML pipelines, statistical modeling, data engineering, responsible AI

**B.Tech Computer Science & Engineering** — JNTU Anantapur
2013 – 2017

## TECHNICAL SKILLS

**Agentic AI & LLM:** MCP (Model Context Protocol), LangGraph, LangChain, Multi-Agent Orchestration, ReAct, Chain-of-Thought (CoT), HITL, Agentic Applications, Intelligent Automation, Workflow Automation, Generative AI, LLM Integration (OpenAI, Gemini, Anthropic, Groq, Ollama), Prompt Engineering, RAG Pipelines, Grounding, LLM Evaluation, Responsible AI, Guardrails, FastAPI, ChromaDB

**Engineering:** Python (expert), JavaScript, TypeScript, PHP, SQL, HTML/CSS, REST APIs, JSON, Microservices, Docker, CI/CD (GitHub Actions), AWS (S3, EC2, RDS, Glue), RabbitMQ, PostgreSQL, MySQL, Oracle, Git, Linux, Agile/Scrum; 300+ code reviews, 5+ engineers mentored

**ML & Data:** PyTorch, scikit-learn, XGBoost, spaCy, BERT, NLP, Hugging Face, NumPy, Pandas, Tableau, Power BI, Statistical Modeling, Anomaly Detection